

Department of Mathematics and Applied Mathematics

3DM: Discrete Mathematics

Test 2

0800 Friday 10 July – 0800 Saturday 11 July 2020

50 marks = 100%



- This question paper consists of **10 pages** (including this one).
- You must upload your solution to the Assignments Tool in the 3DM Vula no later than 8am Saturday 11 July.
- **I shall add 2% to your Test 2 Mark if your submission is a single PDF with file size at most 2MB.**
- Please put your name and student number on your solution.
- In Question 1, you must just give the answer (i.e., no proofs/working necessary). In Question 2, you must show all working/proofs (no working/proof = no marks, even if your answer is correct).
- You may type your solution (I strongly suggest if you do that you use $\text{\LaTeX}$) or (if you have a tablet) write on this question paper and upload the PDF.
- Or you may write your solution out and upload photographs/scans of the pages. If you do this, please make sure each page is clearly legible. **If the photograph is not clear or if we can't read your handwriting, we will mark your solution incorrect.**
- The mathematics in this test should only take 1 – 2 hours to finish. The rest of the time is to accommodate difficulties connecting etc. You should therefore finish the test as soon as possible and upload your solution as soon as you are finished. **Don't wait till the last minute to upload your solution!**

OPEN BOOK TEST RULES

This is an Open Book test. That means:

- You may refer to the course notes and your homework answers.
- You may not use the internet to get answers or help. I had to think about this one for a while. Not everyone in the class can easily access the internet. Allowing students to cruise around the internet looking for things to help them with a test they're busy writing risks creating a very uneven playing field.
- You may not communicate with anyone about the questions (that includes other students in the course, people on the internet, your grandmother the maths professor, your friend who did the course 3 years ago, etc).

Please read the Faculty of Science Student Honour Code, below.

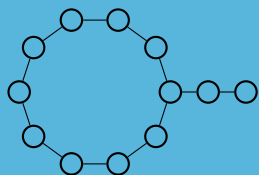
FACULTY OF SCIENCE STUDENT HONOUR CODE

The University of Cape Town's official statement of general rules and policies, including a statement of values and expected student and staff conduct, can be found [here](#). All staff and students in the Faculty of Science are required to familiarise themselves with UCT's rules and policies. The Faculty of Science takes UCT's statement of values seriously. Among other things, every member of the Faculty of Science — student and staff — is expected to strive for personal and academic integrity; to be respectful and tolerant towards others; to honour the rights, personal choices and property of others; to take responsibility for their behaviour individually and in groups; and generally to act as a responsible citizen in our academic community and beyond. As stated in the above mentioned document: "Truth, fairness, consistency and integrity in both academic and other work, and in all personal and institutional relationships" are key elements of UCT's academic values. We are currently in a global crisis, and recognise that distance learning will be a challenge for many students. Nonetheless, we must strive to uphold these values. We ask students to commit to an Honour Code — i.e. doing what is morally right — when completing tasks remotely. In particular, all assignments and tests must be a student's own work, with no input from others (unless otherwise specified), and no work can be plagiarised. We also expect students to report to the Faculty if they believe that other UCT students have breached this code. The Faculty of Science supports our students in this difficult time, and requests their full commitment to the integrity of the academic project. Together we will persevere, and succeed.

Question 1. [15 points] Give an example of each of the following (if such a graph exists) or write Does Not Exist (if it doesn't). **Do not show working!**

- (i) A graph of radius 5 and diameter 7.

(3)



- (ii) A connected graph that has a bridge but no cut-vertex.

(3)



- (iii) A connected graph that has a cut-vertex but no bridge.

(3)



- (iv) A bipartite graph G of order 12 such that for every pair u, v of nonadjacent vertices, the graph obtained from G by adding an edge between u and v is not bipartite.

(3)

$K_{6,6}$

- (v) A tree of diameter 10 having exactly three vertices of eccentricity 5.

(3)

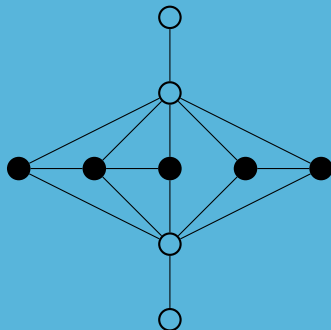
Does Not Exist

Question 2. [35 points] Show all working/proofs!

- a. Prove or disprove: There exists a graph with center $P_3 \cup P_2$.

(5)

This is true. Here is one such graph:



- b. For a positive integer k , a graph is k -partite if there exists a partition of $V(G)$ into k sets $V_1, V_2, \dots, V_k$ such that for each $i \in [1, k]$, the graph $G[V_i]$ is empty (so a 2-partite graph is the same thing as a bipartite graph). Prove or disprove: If G is k -partite, then G does not contain a cycle of length $k + 1$. (5)

This is false. As a counterexample, consider the graph $C_4 : v_1, v_2, v_3, v_4, v_1$. This graph is 3-partite (for example, let $V_1 = \{v_1, v_3\}$, $V_2 = \{v_2\}$, $V_3 = \{v_4\}$), but contains a 4-cycle.

- c. If G is a graph of order n with $\delta(G) \geq \frac{n-1}{2}$, prove that G is connected. (5)

Let k be the number of components of G . Each component contains at least $\delta(G)+1 = \frac{n+1}{2}$ vertices, so $k \frac{n+1}{2} \leq n$, implying that

$$k \leq 2 \cdot \frac{n}{n+1} < 2,$$

implying that $k = 1$, i.e., G is connected.

d. Suppose G is a connected graph and $u \in V(G)$.

- (i) Prove or disprove: If u is a cut-vertex, then $G[N(u)]$ is disconnected. (3)

Suppose u is a cut-vertex and, to the contrary, that $G[N(u)]$ is connected. By Exercise 3.7, u has neighbours x and y such that u is on every $x - y$ path. However, since $G[N(u)]$ is connected, there is an $x - y$ path that does not contain u , a contradiction.

- (ii) Prove or disprove: If $G[N(u)]$ is disconnected, then u is a cut-vertex. (3)

This is false. As a counterexample, let u be any vertex of C_4 . Then $G[N(u)]$ is disconnected, but u is not a cut-vertex.

- e. Let T_1 and T_2 be two vertex disjoint trees. Let T be the graph obtained from T_1 and T_2 by choosing a vertex v_1 of T_1 and a vertex v_2 of T_2 , then adding a new edge v_1v_2 . In other words, $V(T) = V(T_1) \cup V(T_2)$ and $E(T) = E(T_1) \cup E(T_2) \cup \{v_1v_2\}$. Prove that T is a tree. (5)

Since T_1 and T_2 are connected, clearly T is connected. Furthermore, T has order $n(T_1) + n(T_2)$ and size $(n(T_1) - 1) + (n(T_2) - 1) + 1 = n(T) - 1$. Thus by Exercise 4.9(c), T is a tree.

f. Suppose that T is a tree of order $n \geq 2$.

(i) For each positive integer i , let n_i denote the number of vertices of degree i in T . Prove that

$$n_1 = \sum_{i \geq 3} (i - 2)n_i + 2.$$

Since in a tree of order n and size m , we have $m = n - 1$, and since $n \geq 2$, it follows from the First Theorem of Graph Theory that

$$\sum_{i \geq 1} i n_i = 2(n - 1) = 2 \sum_{i \geq 1} n_i - 2,$$

so

$$\sum_{i \geq 1} (i - 2)n_i + 2 = 0,$$

and hence

$$(-1)n_1 + 0 \cdot n_2 + \sum_{i \geq 3} (i - 2)n_i + 2 = 0$$

which after some rearranging becomes the desired result.

(6)

- (ii) Use the preceding result to prove that every nontrivial tree T contains at least $\Delta(T)$ end-vertices. (3)

The number of endvertices is n_1 . By the preceding result,

$$n_1 = \sum_{i \geq 3} (i - 2)n_i + 2 \geq (\Delta - 2)n_\Delta + 2 \geq \Delta$$

since $n_\Delta \geq 1$.